Sentiment Analysis of Drug Reviews Using Machine Learning and Deep Learning Models

Ahmet Furkan Kocabas
Department of Computer Engineering,
Abdullah Gul University
Kayseri, Turkey
ahmetfurkan.kocabas@agu.edu.tr

Erdem Yavuz Hacisoftaoglu
Department of Computer Engineering,
Abdullah Gul University
Kayseri, Turkey
erdemyavuz.hacisoftaoglu@agu.edu.tr

***Abstract*—This study explores sentiment analysis for drug reviews by comparing traditional machine learning, deep learning models, and Transformer Models such as BERT. The dataset, sourced from drugs.com, underwent preprocessing and feature extraction using TF-IDF, Word2Vec, and BERT embeddings. Traditional models such as Logistic Regression, Random Forest, Naive Bayes, and Gradient Boosting were evaluated with TF-IDF and Deep Learning models like CNN and LSTM with Word2Vec for binary classification (positive vs. negative), while BERT was specifically fine-tuned for both binary and three-class (positive, negative, neutral) classification. Results show that BERT achieved an F1-score of 92% in binary classification and 81% in three-class classification, outperforming all other methods. This research highlights the strength of transformer-based models in handling nuanced sentiment tasks in healthcare.**

Keywords: Sentiment analysis, drug reviews, machine learning, deep learning, BERT, NLP, healthcare.

I. Introduction

Patient-generated reviews about medications represent a critical source of information about drug efficacy, side effects, and patient satisfaction. Sentiment analysis is the transformation of these qualitative reviews into quantitative data that healthcare providers and researchers can use to systematically evaluate public perceptions about drugs.

Sentiment analysis has gained significance across various domains like e-commerce and healthcare. Transformer-based models like BERT have further revolutionized NLP tasks, as demonstrated by Devlin et al. [1]. These models are particularly effective in capturing contextual details in text, which makes them suitable for medical sentiment analysis.

The models performed quite well, ranging from traditional methods such as Logistic Regression, Random Forest, CNN, and LSTM, for binary classifications. However, such models falter when nuanced context forms the task at hand-for example, classification of neutral sentiments alongside the positive and negative classes. BERT is a transformer-based model that does exceptionally well on such tasks by leveraging contextual embeddings and pre-training on large corpus. This work focuses on the benchmarking of traditional models in binary classification, while assessing the performance of BERT in both binary and three-class classifications.

II. Background & Related Work

Sentiment analysis has been a focus of NLP research across various domains such as e-commerce, social media, and healthcare. Traditional approaches rely heavily on feature extraction techniques like TF-IDF and bag-of-words, which may provide simple yet effective representations of text. However, these methods fail to address polysemy and contextual semantics, leading to suboptimal performance.

Deep learning introduced word embeddings like Word2Vec, which represent semantic relationships between words in dense vector spaces. LSTMs extended this capability to sequential data, capturing dependencies across time. The transformer architecture, particularly BERT, further advanced NLP by providing bidirectional contextual embeddings. Despite its success in general sentiment tasks, BERT's potential in the healthcare domain, especially for drug reviews, could be further measured. This study aims to address it by evaluating BERT's performance in three-class classification tasks, alongside traditional models for binary tasks.

III. Methodology

A. Dataset

The dataset used in this study was sourced from drugs.com, a popular online platform where patients share their experiences with various medications [2]. It consists of 161,297 samples in the training set and 53,766 samples in the test set. Each review is labeled from 1 to 10 inclusive, reflecting the patient's sentiment about the drug's effectiveness and side effects.

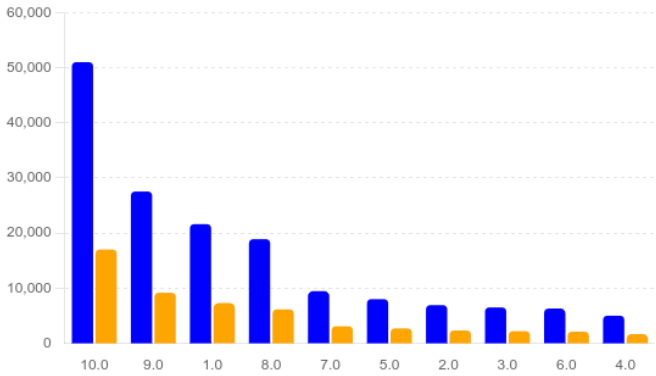

Figure 1: Train and Test Set Class Distribution

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©2025 IEEE

Despite its size, the dataset exhibits class imbalance, with the majority of reviews being labeled as 10, 9 and 1, which presents a significant challenge for classification models. Hence, a class mapping was applied to the datasets to tackle the class imbalance. For three-class classification, random under sampling was further applied to balance the three classes.

	Negative	Neutral	Positive
Binary	0-6	-	7-10
Three-Class	1-3	4-7	8-10

Table 1: Class Mappings

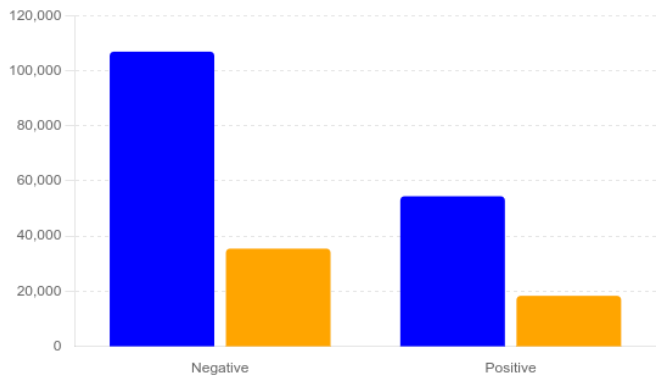

Figure 2: Binary Class Distribution of Train and Test Set after Class Mappings

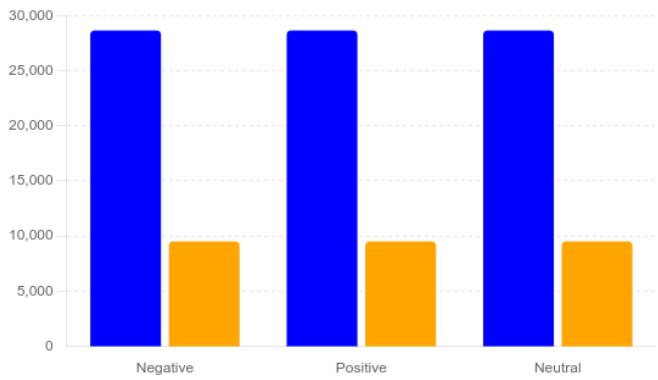

Figure 3: Three Class Distribution of Train and Test Set after Class Mappings and Undersampling

By applying random under sampling and class mappings for three-class classification, the dataset was rebalanced, enabling models like BERT to learn effectively across all sentiment categories

B. *Preprocessing*

To ensure the data was ready for analysis, a series of preprocessing steps were applied: The reviews were first cleaned by removing HTML tags, URLs, and special characters, which often contribute noise to textual data. Subsequently, all text was normalized by converting it to lowercase, ensuring uniformity across the dataset. Tokenization was applied to split sentences into words, and stop words were removed to reduce noise further. Rare words and non-alphanumeric tokens, which hold little significance in sentiment analysis, were also excluded. Lastly, class labels were standardized, and the Minority Undersampling technique was employed to address the dataset's inherent imbalance.

C. *Feature Extraction Methodologies*

Three distinct feature extraction techniques were utilized, each suited to different types of models:

TF-IDF (Term Frequency-Inverse Document Frequency): This traditional method was used to represent text as numerical vectors, capturing the importance of words within individual reviews relative to the entire dataset. It served as the primary input for simpler machine-learning models.

TF-IDF has been a essential approach for text classification tasks owing to its ability of capturing significance of words in document [2]. Hence, it is utilized to extract features from drug reviews to train machine learning models in this study.

Word2Vec: The tool created word embeddings with high density, featuring semantic relationships among words. The text embeddings obtained represented richer information, ideal for deep learning models such as LSTM and CNN.

BERT (Bidirectional Encoder Representations from Transformers): As a transformer-based model, BERT generates contextualized embeddings by considering both the preceding and succeeding words in a sentence. This advanced feature representation enabled models to capture subtle semantic nuances, which are critical in healthcare text.

D. *Models*

This study evaluated a range of traditional and deep learning models to compare their effectiveness:

Traditional Machine Learning Models: Naive Bayes, Logistic Regression, Random Forest, and Gradient Boosting were implemented as baseline models. These algorithms are known for their computational efficiency and interoperability.

Deep Learning Models: Advanced architectures like LSTM, CNN, and BERT were deployed to leverage sequential and contextual data representation. LSTM focused on learning long-term dependencies, CNN extracted local patterns, and BERT captured deep contextual meanings with its transformer architecture.

Binary Classification: Logistic Regression, Naive Bayes, Random Forest, Gradient Boosting, LSTM, CNN,BERT

Three-Class Classification: BERT

E. *Evaluation Metrics*

To assess the performance of each model, we used a comprehensive set of metrics:

Accuracy: A basic measure of the proportion of correct predictions.

Precision: Evaluates the fraction of correctly identified positive instances among all instances classified as positive.

Recall: Measures the model's ability to detect all actual positive instances.
F1-Score: A harmonic mean of precision and recall, balancing false positives and false negatives. Additionally, confusion matrices were analyzed to understand the distribution of true positives, false positives, and misclassifications for each model.

IV. Result & discussion

A. TF-IDF with Traditional ML Models Performance

Traditional machine learning models served as a baseline for sentiment analysis. Various machine learning models were evaluated using TF-IDF representations with up to 5,000 features, including bi-gram windows Among these, Random Forest and Logistic Regression seemed perform better than others.

	Precision	Recall	F1	Accuracy
Naive Bayes	0.78	0.78	0.77	0.78
Logistic Regression	0.81	0.82	0.81	0.82
Random Forest	0.81	0.81	0.79	0.81
Gradient Boosting	0.75	0.74	0.70	0.74

Table 2: TF-IDF + Machine Learning Models' Performance

These results highlight that TF-IDF as feature extraction technique with traditional machine learning approaches could end up with promising results, over .70 as F1 score.

B. Word2Vec with Deep Learning Models Performance

Deep learning models demonstrated superior performance compared to traditional methods. LSTM achieved an F1-score of 83%, effectively capturing sequential patterns in the text. However, CNN, while faster to train, lagged behind in handling contextual dependencies compared to TF-IDF + Logistic Regression and Random Forests. The probable reasons will be further discussed in the later sections.

	Precision	Recall	F1	Accuracy
LSTM	0.83	0.82	0.83	0.83
CNN	0.78	0.78	0.78	0.78

Table 3: Word2Vec + Deep Learning Models' Performance

C. BERT: Binary and Three-Class Classification:

An already pretrained BERT model on large lowercase English text corpus, utilizing transformer-based contextual embeddings, was evaluated for both binary and three-class sentiment classification tasks. The results are as following:

	Precision	Recall	F1	Accuracy
Binary	0.91	0.91	0.91	0.91
Three-Class	0.84	0.81	0.82	0.81

It turns out that BERT achieved state-of-the-art results with an F1 score of .92, outperforming all other combinations of models and representations as expected. Moreover, BERT could yield an F1 score of .80 in three class classification implying that BERT could able to capture semantic meanings of reviews even the review does not imply direct sentiments. For example, "Not bad" labeled as positive in binary classification but neutral in three-class classification which is more convenient.

D. BERT's Limitations:

While BERT demonstrates exceptional performance in sentiment analysis, there are certain limitations that result in misclassifications, as evident from specific review examples. These limitations highlight the complexity of natural language and the challenges associated with understanding subtle context, sarcasm, or nuanced sentiment.
Misclassification of a Negative Sentiment Review Text:

"I am a 12-year survivor of mirapex/Pramepexole. It was a miracle to begin with. I only slept about 4 hrs a night but had enormous energy and never tired. That lasted nine years. But during that time I lost bladder control, impulse disorders, loss of concentration, myalgia neuropathy, falling asleep while driving, many more and augmentation. I was on .5 mg. Detoxing med-free from this med was horrendous! After detox, most of the side effects cleared up. But I still have issues with memory and concentration. This is the only med I took on a reg basis as my health was good. I'm still suffering the side effects of the Withdrawal from it. Studies should be for longer than 12 weeks."

True Label: Negative
Predicted Label: Positive

The review contains a mix of initial positivity and a detailed account of adverse effects. BERT likely misclassified this as positive due to the early statement, "It was a miracle to begin with," which sets a positive tone. However, the predominant sentiment is negative, with severe side effects and ongoing issues emphasized later in the review. This could mean that BERT's transformer-based attention may overly prioritize earlier positive phrases while failing to sufficiently weigh subsequent negative details.

A concise way to classify reviews that mix positive impressions (e.g., "miracle" results) with negative aspects (e.g., severe side effects) might be to adopt **multi-aspect sentiment analysis**. Specifically, identifying and classifying significant aspects of the review, such as the drug's initial effectiveness versus its long-term impact and withdrawal issues, rather than assigning a single overall sentiment. This approach could have prevented a positive comment from overshadowing later negative details and nuances, ensuring a more accurate evaluation of the patient's sentiments.

V. Conclusion

Overall, this study aimed to evaluate both traditional and deep learning approaches for sentiment analysis on the drugs.com dataset, applying binary and three-class classification tasks. Surprisingly, simpler methods, such as Logistic Regression with TF-IDF, often matched or outperformed deep learning models such as CNN with Word2Vec, likely due to the dataset's narrow vocabulary and straightforward narratives. Nonetheless, BERT exhibited a clear advantage in binary classification, showcasing superior contextual understanding, and also performed strongly in three-class classification by effectively capturing mixed sentiments. However, even

BERT showed weaknesses while dealing with contradictory or complex drug reviews.

ACKNOWLEDGMENT

This study was conducted as part term project in ECE 544 “Text Mining” course at Abdullah Gül University during the Fall 2024 semester. We would like to extend our sincere gratitude to **Asst. Prof. Mehmet Gökhan Bakal** for his invaluable guidance, support, and expertise throughout the project.

REFERENCES

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in Proceedings of NAACL-HLT 2019, Minneapolis, MN, USA, Jun. 2019.

[2] R. El Kharoua, “Patient Ratings Identifying Best Drugs,” Kaggle, Accessed: Oct. 12, 2025. [Online]. Available: https://www.kaggle.com/datasets/rabieelkharoua/patient-ratings-identifying-best-drugs/data

[3] J. Ramos, "Using TF-IDF to Determine Word Relevance in Document Queries," in Proceedings of the First International Conference on Machine Learning, New Brunswick, NJ, USA, Nov. 2003.